Electromagnetic induction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why changing magnetic flux can induce an emf.

SUCCESS CRITERIA - I can...

1. Identify a change in magnetic flux.
2. Predict whether a stationary arrangement produces sustained induction.
3. Use Lenz's law to describe opposition to change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "induction" and "flux" to explain your model or result. Sentence starters are optional.

induction: producing an emf through a changing magnetic flux

flux: a measure of magnetic field passing through a chosen area

emf: energy supplied per charge, measured in volts

01 Warm-up: bring back an earlier idea

A 3 W device runs for 5 s.
Find transferred energy.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

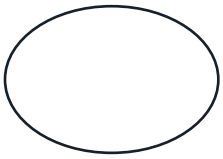
Reference: $\Phi = BA$; $|emf_{avg}| = N|\Delta\Phi|/\Delta t$.
Uniform perpendicular field for $\Phi = BA$. N is coil turns; the induced effect opposes the flux change.

Worked reasoning: A changing magnetic flux through a coil can induce voltage. Holding a magnet stationary relative to a fixed coil gives no induction from relative motion in this simple setup.

Your turn: Draw a magnet approaching a coil and mark the changing field through the loop. Contrast it with a stationary case.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

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Continue down each column.

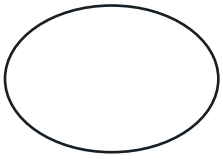
Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Compare a magnet held still beside a fixed coil with a magnet moving into the coil. Which can produce a changing flux?

Sentence starter: Both … . They differ in … because … .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

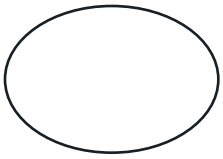
Marzano check - circle: ① ② ③

04 Apply the physics

A lab group pushes a magnet into a fixed coil and records a positive meter deflection.
Draw the approach and withdrawal motions.
Predict the sign when the magnet is pulled back out along the same path without changing connections.

Sentence starter: I predict … . My model supports this because … .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

05 Explain your reasoning

Why does Lenz’s law oppose the change in flux rather than always oppose the field itself?

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Keep charge separation fixed and multiply one charge magnitude by 4.
Compare electric-force magnitude.
Label charges and both interaction forces.

Word help: separation = distance between the specified objects or centers

Sentence starter: Both … . They differ in … because … .



Charge A Charge B

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

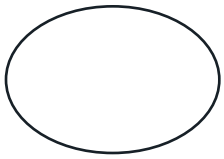
07 Exit check

Name two ways to increase the magnitude of induced emf in a simple magnet-and-coil trial.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I notice … . This supports … because … .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

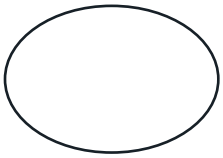
Marzano check - circle: 1 2 3

08 STRETCH: test the model

Predict how faster approach and more turns can change induced voltage.
State which comparisons must keep the other conditions fixed.

Sentence starter: I predict … . My model supports this because … .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3